

Building Open and Reproducible Geospatial Simulations with Python

DisSModel — Discrete Spatial Modeling Framework

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Cellular Automata

System Dynamics

LUCC · Mangroves

LUCC · CLUE

AGENDA

What we will cover today

01 Contextualization

3 min

02 The Problem

6 min

03 DisSModel: A Discrete Spatial
Modeling Framework for Python

5 min

04 The DisSModel Architecture

5 min

05 Laboratory Models

3 min

06 Empirical Models

4 min

07 What Do We Gain with DisSModel?

2 min

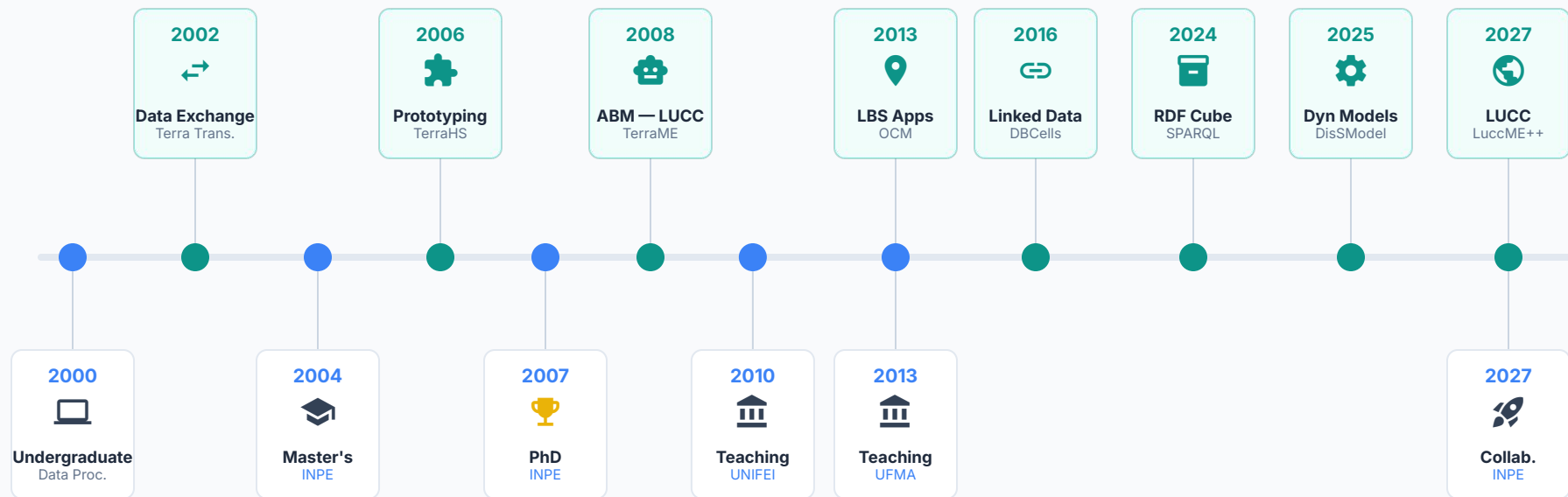
08 Next Steps

2 min

Career and Research Evolution

● Research Projects

● Academic & Professional



Focus on Spatial Modeling and Geographic Data (Open Science)

Geospatial Simulations

Land Use and Cover Change (LUCC)

LUCC models how and why the land surface changes over time. It is the mathematical language of human consequences on the territory.

Deforestation

Conversion of native forest to agricultural use.

Coastal Dynamics

Mangroves vs. urbanization and sea level rise.

Urban Expansion

Growth of cities over rural and natural areas.

Agricultural Frontier

Agribusiness advance over the Cerrado and the Amazon.

FOCUS: PROCESS-BASED GEOSPATIAL MODELING

Expectation Alignment

This presentation focuses exclusively on **Geospatial Models** instead of general-purpose Machine Learning models.

Key Brazilian Geospatial Tools

Dinamica EGO

UFMG · since 2002

Visual GUI platform for spatially explicit LUCC modeling.

STRENGTHS

- No-code visual interface
- Academic benchmark

LIMITATIONS

- Manual reproducibility
- Scalability issues

TerraME

INPE · since 2005

Lua simulation framework for cellular automata models.

STRENGTHS

- Native Agent/Space logic
- PostGIS integration

LIMITATIONS

- TerraLib is no longer supported
- No BDC/STAC support

LuccME

INPE · built on TerraME

Modular framework based on Demand/Allocation architecture.

STRENGTHS

- Interchangeable parts
- CLUE methodology

LIMITATIONS

- Inherits Lua constraints
- Weak Python link

The reproducibility crisis in computational science

Baker (2016), Nature · NAS (2019) · Stodden et al. (2016), Science · Kravtsov et al. (2021)

Baker, Nature 2016

70%

of researchers could not reproduce experiments from other groups.

NAS, 2019

National Academies, commissioned by the U.S. Congress. Formal report defining computational reproducibility as a pillar of modern science and establishing institutional recommendations.

Code doesn't run

Undocumented dependencies, incompatible versions, **implicit environments** — the experiment only works on the author's machine.

Inaccessible data

Input data not versioned or not published. **Without original data**, the result cannot be audited or replicated.

Opaque parameters

Critical model configurations are in ad hoc scripts or in the **researcher's memory**, not in formal auditable records.

Challenges in Dynamic Geospatial Modeling

Spatial simulations have additional challenges beyond general computational reproducibility

01

Geospatial data is large and volatile

Rasters and vectors with specific CRS, varying resolutions and version history. A different GeoTIFF silently masks calibration errors.

02

Models couple science and infrastructure

The CA or LUCC logic gets mixed with file I/O, visualization and loop control — impossible to reuse without rewriting.

03

Experiments are not traceable

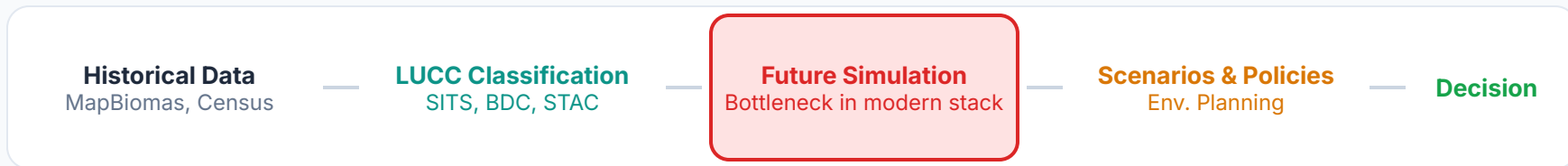
Which model version? Which parameters? Which input dataset? Without formal records, the result is irreproducible by design.

04

Scale breaks the scientific workflow

The researcher calibrates in the notebook, but running scenarios at scale requires rewriting the code — or abandoning production execution.

Why doesn't LambdaGEO propose modernizing TerraME?



	TerraME / LuccME	DisSModel (The Proposal)
Language	Lua + TerraLib (no longer maintained)	Pure Python (Geopandas, NumPy)
Reproducibility	No formal record	Git + SHA-256 per execution
Cloud Integration	Isolated local execution	Cloud Native (API, Minio, Docker)

DisSModel preserves the TerraME modeling philosophy in a modern ecosystem.

The Researcher's Journey: End-to-End Workflow

Modules of the @dissmodel Organization

PHASE 1 DEVELOPMENT & PACKAGING

1. Creation

Model design using the base library (model, environments, geo models, rules).

2. Packaging

Logic is fully encapsulated in *executors* (validate, load, execute, save).

3. Tests

The model is validated locally in an incredibly agile way via CLI tool.

4. Publishing

Library source code made available in independent GitHub repositories.

PHASE 2 INTEGRATION & EXECUTION

5. Integration

Researcher creates a .toml manifest and submits a PR to the catalog.

6. Curation

Maintainers review compliance and security before official incorporation.

7. API Execution

Platform manages dependencies and exposes results via dynamic API.

"Science should not need to be rewritten to go into production." — DisSModel Principle

Core Principles

The fundamental pillars for robust and scalable scientific modeling.

Reproducible Research

Data & code sharing for third-party verification.

IMPLEMENTATION:

- Executor + TOML + Docker
- Auditable Record (SHA-256)
- Code/Parameter separation

Claerbout (1992) · Victoria Stodden (2016), Science

FAIR for Software

Findable, Accessible, Interoperable, and Reusable.

KEY PILLARS:

- F:** DOI + Metadata
- A:** API + S3/MinIO
- I:** GeoTIFF + Xarray
- R:** TOML Versioning

Wilkinson et al., 2016

Software Engineering

Bridging the gap: Scientists vs. Production Engineers.

LAYER SEPARATION:

- Science:** logic is kept pure.
- Platform:** APIs & infrastructure.
- SimOps:** Scientific SimOps.

Baxter et al. (2012) · Wilson (2017) PLOS Comp. Bio.

"Science should not need to be rewritten to go into production." — DisSModel Principle

Main Repositories

Modules of the @dissmodel Organization



Base

3 repositories

Provides the model's core engine, platform infrastructure, and official catalog.



Lab Models

2 repositories

Focused on cellular automata and dynamic systems for testing, teaching, and learning.



Empirical Models

2 repositories

High-fidelity models previously implemented in TerraME, now available as reproducible Python workflows.



Interfaces

1 repository

A QGIS plugin that runs the BR-MANGUE model natively — no API calls, no middleware, simulation inside the GIS environment.

Base Components

dissmodel

LIBRARY (Core Framework)

Central simulation engine for local scientific model development and execution.

github.com/DisSModel/dissmodel

```
pip install dissmodel
```

dissmodel-platform

PLATFORM (Cloud-Native)

Open and scalable infrastructure for distributed cloud experiment execution.

github.com/DisSModel/...-platform

```
docker compose up --build
```

dissmodel-configs

MODEL CATALOG

The single source of truth. Centralizes configuration files (manifests in .toml).

github.com/DisSModel/...-configs

```
pull requests
```

*"Science should not need to be rewritten to go into production" — **dissmodel** runs the same code locally and on the remote **platform**.*

Core Framework

core

Simulation Engine

Discrete clock +
execution cycle via
Environment & Model.

*Environment, Event
dispatch, step control.*

geo

Spatial Substrate

Dual design: Vector
(GeoDataFrame) +
Raster (NumPy).

*Queen/Rook and
Moore/Von Neumann
neighborhoods.*

executor

Execution Layer

ModelExecutor ABC +
SimulationRegistry.

*Local CLI and Remote
Workers — same code.*

io

Storage I/O

Local file and MinIO/S3
read/write operations.

*Checksums and
provenance.*

vis

Visual Observers

Chart, Map,
RasterMap,
display_inputs.

*Decoupled Observer
pattern.*

"Science should not need to be rewritten to go into production" — **DisSModel Principle**

DisSmodel-platform - Server Side

FastAPI Gateway

API-Driven Execution

- Programmatic REST calls
- Legacy dashboard integration
- Validation via Pydantic

Docker Workers

Subprocess Isolation

- Jobs in isolated containers
- No dependency conflicts
- Redis Queue scalability

Jupyter Lab

Experiment Service

- Interactive experimentation
- Direct data visualization
- Rapid model prototyping

MinIO Storage

Agnostic Storage

- S3-compatible integration
- Unified spatial writing
- Zarr and GeoTIFF support

*The infrastructure orchestrates **FastAPI, Redis, and Docker** to ensure science does not need to be rewritten for production.*

DisSModel-configs - The Central Catalog:

Single Source of Truth

To keep the ecosystem scalable and secure, we separate the model code from its publication.

The catalog stores only declarative manifests, ensuring full traceability and governance.



Manifestos .toml

Each model has a centralized configuration file.



Versioning

Points to the specific **release** or **tag** on GitHub.



Parameterization

Defines default limits consumed by the API.

```
configs/models/coastal_raster.toml

[model]
executor_module = "br-mangue.executor"
name = "br-mangue"
package = "git+https://github.com ..."

[model.parameters]
end_time = 88
taxa_elevacao = 0.5
altura_mare = 6.0
acrecao_ativa = false
resolution = 100.0
crs = "EPSG:31984"
```

Classical Models (SysDyn and CA)

disssmodel-ca

CELLULAR AUTOMATA



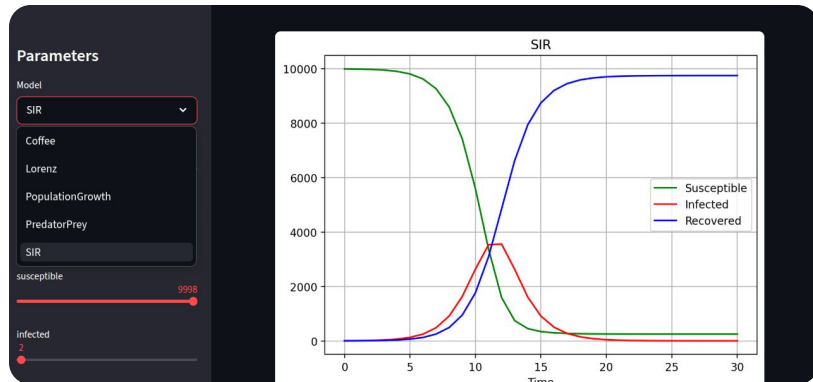
Spatial Modeling

Complex systems through local interactions in spatial grids.

- Game of Life & Forest Fire
- Substrate: GeoDataFrame
- Processing via NumPy

disssmodel-sysdyn

SYSTEM DYNAMICS



Temporal Dynamics

Focus on flows, stocks, and feedback loops for temporal behavior.

- SIR, Lorenz, Predator-Prey
- Streamlit Explorers
- CLI Interface

Cellular Automaton: Game of Life

Model Definition

```
class GameOfLife(CellularAutomaton):
    def setup(self) → None:
        self.create_neighborhood(strategy=Queen)

    def rule(self, idx) → int:
        state = self.gdf.loc[idx, 'state']
        n = self.neighbor_values(idx).sum()
        if state == 1:
            return 1 if 2 ≤ n ≤ 3 else 0
        return 1 if n == 3 else 0
```

Model Execution

```
gdf = vector_grid((20, 20), attrs={"state": 0})
env = Environment(0, 10)

GameOfLife(gdf).initialize()

Map(gdf)

env.run()
```

Dynamic Systems: SIR Model



Model Definition

```
class SIR:
    def execute(self):
        total = S + I + R
        alpha = contacts * prob
        new_inf = I * alpha * (S/total)
        new_rec = I / duration
        S -= new_inf
        I += new_inf - new_rec
        R += new_rec
```

Model Execution

```
env = Environment()
SIR(9998, 2, 0, 2, 6, 0.25)
Chart("SIR Model")
env.run(30)
```

DisSModel in Real-World Scenarios

brmangue-dissmodel

REFERENCE

Coastal Case Study

- Flooding and Mangroves
- Based on Bezerra et al. (2014)
- Focus Area: Maranhão Coast

DisSLUCC-Cont.

REFERENCE

Land Use and Cover

- Demand + Potential + Allocation
- CLUE and LUCCE Structure
- INPE/Brazil Methodology and Expertise

Coastal Case Study (brmangue)



Study Area

Maranhão Coast, Brazil. Vector spatial grid.



Scientific Basis

Bezerra et al. (2014).



FloodModel

Flooding due to sea level rise scenarios.



MangroveModel

Migration by tide and soil type dynamics.



```
env = Environment(start_time=2008, end_time=2030)
# Add specialized models
FloodModel(gdf=gdf, elevation_rate=0.5,
use_attr='usage')
MangroveModel(gdf=gdf, tide_height=6.0,
soil_attr='soil')

env.run()
```

Performance Benchmark

SUBSTRATE | MS/STEP | SPEEDUP

Vector: **1241.7 ms** (1.0x)

Raster: **31.5 ms** (39.4x ⚡)

Accuracy vs TerraME: **99.8% Match**

Coastal Case Study (brmangue)

Plugin Feature

QGIS Integration

Dedicated plugin for the brmangue model, enabling streamlined simulation and visualization of coastal dynamics within the QGIS environment.

- Intuitive interface for parameter input
- Direct visualization of model outputs
- Seamless integration with vector grids

The screenshot displays the QGIS interface with the BR-MANGUE Coastal Simulation Platform plugin. The main map shows a coastal area with a yellow and green simulation output. The plugin window is open, showing the 'Servidor DisModel Platform' configuration, 'Dataset de Entrada' (input dataset), 'Parâmetros do Modelo' (model parameters), and 'Bandas a Visualizar' (bands to visualize). The 'Status' section shows a message about the simulation process. The 'Citação FAIR' (FAIR Citation) section provides a citation for the DisModel v1.0 plugin. The 'Logs de Execução' (Execution Logs) section shows the simulation progress and completion details.

Servidor DisModel Platform

URL:

API Key:

Dataset de Entrada

URI do dataset:

Formato:

Parâmetros do Modelo

End time (steps):

Taxa de elevação:

Altura da maré:

Acreção ativa: ☐

Bandas a Visualizar

☒ uso (uso e cobertura) ☐ solo (tipo de solo) ☐ alt (elevação)

Status

Aguardando submissão.

[] Carregando pré-visualização de s3://dissmodel-inputs/ilha_maranhao_epsg31983.tif...

x Erro ao carregar pré-visualização: O servidor não retornou uma URL válida para download.

[] Carregando pré-visualização de s3://dissmodel-inputs/ilha_maranhao_epsg31983.tif...

Painel de Experimento — 41f50dd0

Citação FAIR

DisModel v1.0 (spec: brmangue@2ef1bbc, output sha256: 964ce4ff1d5c...)

Proveniência e Hashes

Status: COMPLETED

Model Commit: 2ef1bbc

Input SHA256: 1b06982d2dc04c4d8211dc717afa89f76d932e575fbafe90e6af44e774e30c

Output SHA256: 964ce4ff1d5c81d1a625740f4cf1ebb26d00d307f18ab92f2f2bcb235e5dc94e

Resolved Spec (Parâmetros)

Logs de Execução

```
Record created - model=brmangue commit=2ef1bbc
Dispatching to subprocess...
Loaded GeoTIFF: shape=(418, 405) start=1 crs=EPSG:31983
Running steps 1 - 88...
Simulation complete
Saved to s3://dissmodel-outputs/experiments/41f50dd0b3b504a63a9e931c93a2d53e6/output.tif
Saved profiling artifact - s3://dissmodel-outputs/experiments/41f50dd0b3b504a63a9e931c93a2d53e6/profiling_41f50dd0.nd
Saved record JSON - s3://dissmodel-outputs/experiments/41f50dd0b3b504a63a9e931c93a2d53e6/41f50dd0b3b504a63a9e931c93a2d53e6/record.json
Completed - val=0.80s | loaded=15s | run=0.93s | save=0.04s | total=7.12s
```

DisSLUCC-Continuous: Land Use & Cover

Continuous Modeling

The package implements the CLUE-type allocation algorithm (Veldkamp & Fresco, 1996). Cells receive *fractional proportions* instead of discrete classes.

Evaluation Metric

For continuous outputs, the **MAE (Mean Absolute Error)** was adopted to directly evaluate modelling error (Willmott, 2005).

1. Demand

Magnitude of change per timestep.
Pre-computed via historical CSV.

```
DemandPreComputedValues(  
    annual_demand=load_csv( ... )  
)
```

2. Potential

Change suitability via linear
regression with driver variables.

```
PotentialLinearRegression(  
    gdf=gdf, drivers=[ ... ]  
)
```

3. Allocation

CLUE-like distribution based on
potential and constraints.

```
AllocationClueLike(  
    demand, potential  
)
```

Experimental Setup

DATASET
LuccME AC (2008-2014)

TIME STEPS
6 Steps

RESOLUTION
6,574 cells

DisSLUCC - Validation and Performance Results

Performance (6 steps · 6,574 cells)

3.9×

SPEEDUP RASTER VS VECTOR
(31.6 ms vs 122.5 ms)

Accuracy vs TerraME

0.002276

MAE VS TERRAME · BOTH SUBSTRATES
97.09% match · Vector = Raster · MAE 0.002276

Technical Highlight: Raster substrate achieves 3.9× speedup over Vector (31.6 ms vs 122.5 ms/step). Both substrates match TerraME at 97.09% — and produce identical output (MAE = 0.000), confirming the raster optimisation preserves full algorithmic correctness.

Dataset: LuccME Package - AC - 2008–2014 · 6 steps · 6,574 cells · tolerance 0.01

TRANSFORMING GEOSPATIAL SCIENCE



Hidden Engineering

The complex cycle of data orchestration runs seamlessly.



Full Traceability

Guaranteed data provenance for every simulation stage.



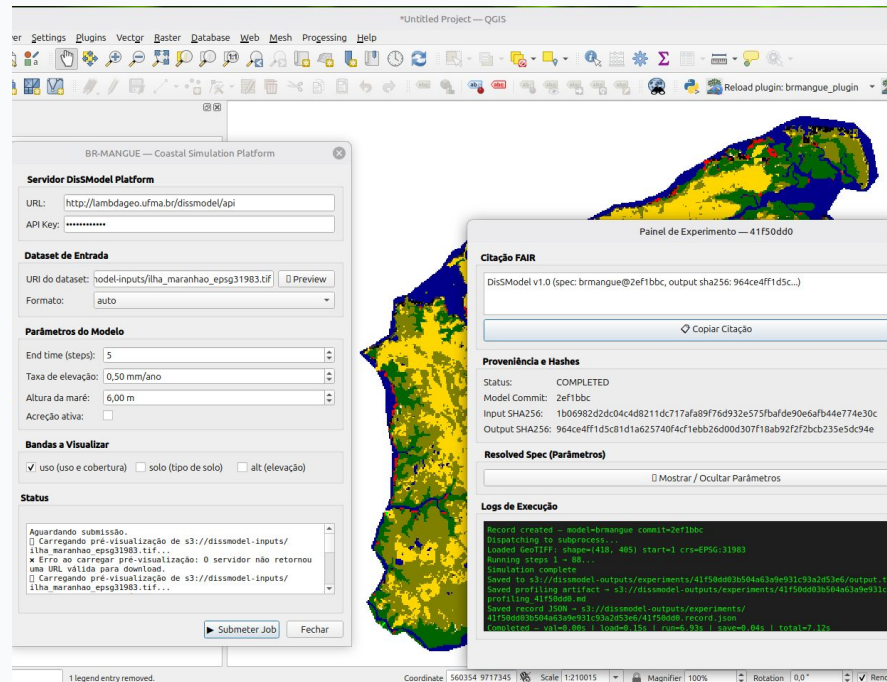
Direct Delivery

Dynamic results delivered directly within GIS tools.



End-to-End Focus

Researchers focus on science, users on decisions.



DisSModel Roadmap 2026–2028

2026 S1

2026 S2

2027 S1

2027 S2

2028

JOSS Publication

Repo organisation &
PyPI v0.5

Formalisation at UFMA

Research plan
submissions

Project Kickoff

CLUE & CLUE-S
(PIBIC)

DisSLUCC (PIBIC)

BRMangue JOSS

DisSModel v1.0

BDC Integration

STAC source reader

Xarray (time, y, x)

ExpRecord → MinIO

Dask / Ensemble

Lazy eval (Pangeo)

Parallel scenarios

Native Zarr output

Book + Community

Geospatial DS book

PT-BR Edition

Workshop INPE/SBSR

* QGIS plugins (qgisparql-triple2layer / layer2triple): joint JOSS submission under evaluation.

Strategic Partnership: DisSModel & LuccME++

2027/28 — DisSModel will serve as the innovation laboratory for the evolution of **LuccME** at CCST/INPE.



DisSModel

Partnership &
Evolution
----->



Future of LUCC

TerraME was the starting point — **DisSModel** advances as the foundation for **LuccME++**

Thank you!

DisSModel — open source, open science

GitHub

`disssmodel.github.io`

Docs

`disssmodel.github.io/disssmodel/`

PyPI

`pip install disssmodel`

Contact

`sergio.costa@ufma.br`

"Science should not need to be rewritten to go into production."
— DisSModel Principle

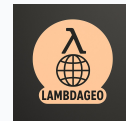
`disssmodel`

`disssmodel-ca`

`disssmodel-sysdyn`

`brmangue-disssmodel`

`DisSLUCC-Cont.`



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Modeling

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